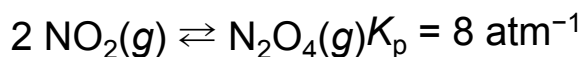


Consider the dimerization of nitrogen dioxide at room temperature:



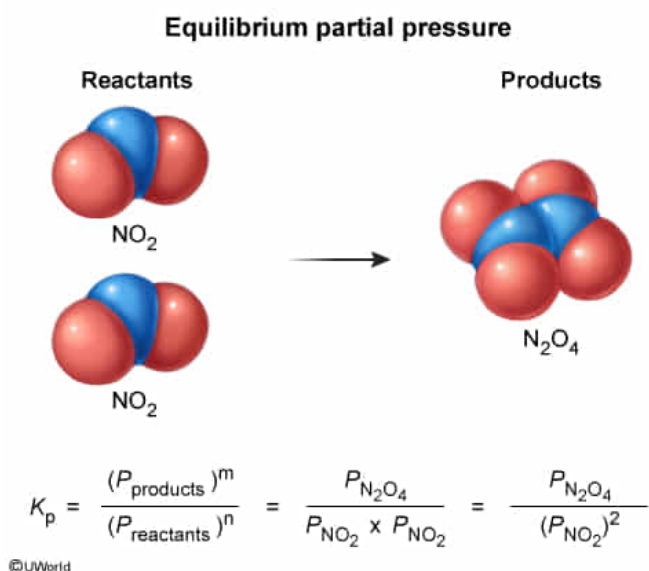
If the equilibrium partial pressure of nitrogen dioxide in a container is 0.5 atm, what is the partial pressure of dinitrogen tetroxide?

- ☐ A. 0.03 atm [10%]
☐ B. 0.5 atm [15%]
☒ C. 2 atm [50%]
☐ D. 4 atm [23%]

Correct

50% answered correctly

Explanation:



The **equilibrium constant** for any chemical reaction is the **molar ratio of products to reactants** when the system is **equilibrium**. The **partial pressure** of a gas is directly proportional to the number of moles of that gas according to the **ideal gas law**; therefore, partial pressure can be substituted for molar concentration. If more than one of a given molecule is involved in a reaction, each molecule involved is counted separately. Therefore, each product or reactant must be **raised to the power** of its **stoichiometric coefficient** to account for all molecules involved in the reaction.

Nitrogen dioxide (NO₂) can combine with itself to form dinitrogen tetroxide

(N_2O_4). The equilibrium constant for this reaction ($K_p = 8 \text{ atm}^{-1}$) is the ratio of the partial pressure of N_2O_4 to that of NO_2 when equilibrium is achieved. However, because two NO_2 molecules are required for each reaction, the NO_2 partial pressure ($P_{\text{NO}_2} = 0.5 \text{ atm}$) must be squared. Therefore, the

equilibrium partial pressures of NO_2 and N_2O_4 in the described scenario are related by the equation:

$$K_p = \frac{P_{\text{N}_2\text{O}_4}}{(P_{\text{NO}_2})^2}$$

Substituting the values into the equation yields:

$$8 \text{ atm}^{-1} = \frac{P_{\text{N}_2\text{O}_4}}{(0.5 \text{ atm})^2} = \frac{P_{\text{N}_2\text{O}_4}}{0.25 \text{ atm}^2}$$

This equation can be rearranged and solved as follows:

$$P_{\text{N}_2\text{O}_4} = 0.25 \text{ atm}^2 \times 8 \text{ atm}^{-1} = 2 \text{ atm}$$

(Choice A) 0.03 atm would be correct if N_2O_4 were the reactant and NO_2 were the product, in which case N_2O_4 would be in the denominator of the equilibrium expression.

(Choice B) 0.5 atm would be correct if "equilibrium" meant that all components are at equal concentration. Instead, equilibrium is the state in which the forward and reverse reactions occur at equal rates and no net change is observed. More stable compounds dominate at equilibrium.

(Choice D) 4 atm would be correct if only one molecule of NO_2 were involved in the reaction and its partial pressure were not squared in the equilibrium expression.

Educational objective:

The equilibrium constant (K_p) of a reaction is the molar ratio of products to reactants when equilibrium is achieved. Each product or reactant must be raised to the power of its stoichiometric coefficient in the equilibrium expression.

Time spent:0 sec

Subject:

General Chemistry

QID:401362

System:

Thermodynamics, Kinetics & Gas Laws